

● EASY

● INFORMATION AND IDEAS

Information and Ideas

14

6 questions · ~9 min

Q1

INFORMATION AND IDEAS

Roasted green chiles are a popular ingredient in Southwestern cuisine, but the traditional roasting method of burning propane is not environmentally friendly. To see if solar power could provide a better alternative, engineer Kenneth Armijo and his team roasted batches of green chiles using between 38 and 42 heliostats, which are devices that concentrate sunlight. The team was successful in reaching the same roasting temperature used in traditional propane roasting, but they found that propane yielded faster results. While the fastest solar-roasted green chiles took six minutes, batches using propane took only four. Armijo hypothesizes that they can reduce the roasting time for solar-roasted green chiles by using more heliostats.

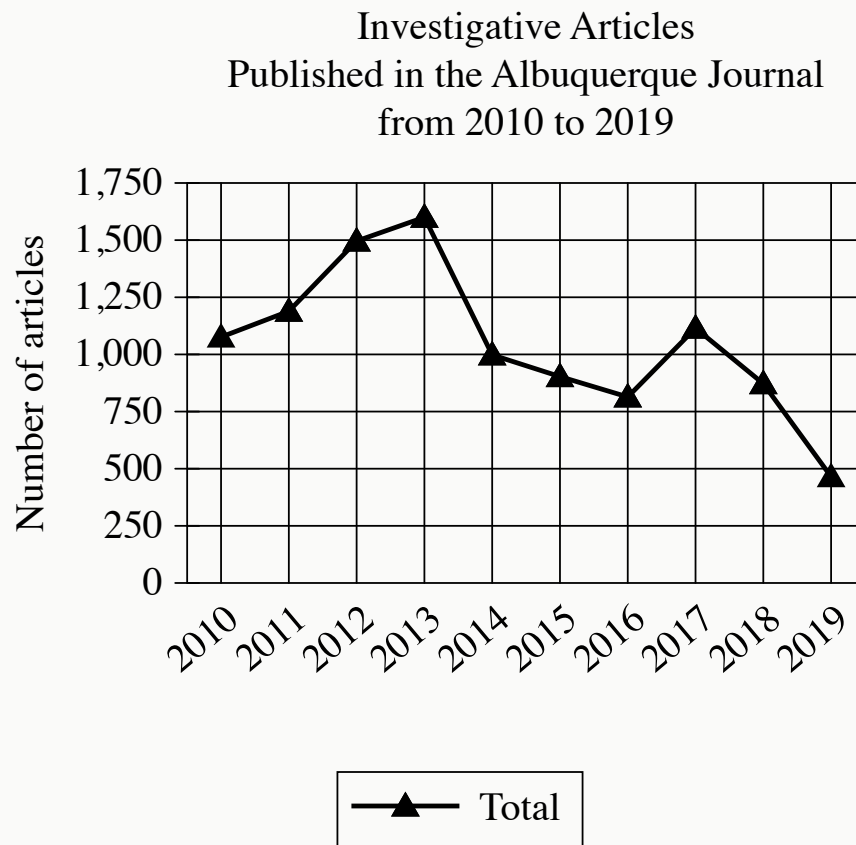
Which finding, if true, would most directly support Armijo's hypothesis?

- A. The temperature inside the roasting drum is distributed more evenly when roasting green chiles with solar power than with propane.
- B. Attempts to roast green chiles using 50 heliostats yields results in fewer than six minutes.
- C. Green chile connoisseurs prefer the flavor of solar-roasted green chiles over the flavor of propane-roasted green chiles.
- D. The skins of solar-roasted green chiles are easier to peel than the skins of propane-roasted green chiles.

“We Are Marching” is a 1921 poem by Carrie Law Morgan Figgs. In the poem, the speaker predicts future success: _____ blank

Which quotation from “We Are Marching” most effectively illustrates the claim?

- A. “Can’t you hear the sound of feet?”
- B. “You who are out just get in line.”
- C. “We have answered duty’s call.”
- D. “We shall never know defeat.”



- The line graph:
 - Begins at 2010, 1,076
 - Rises gradually to 2011, 1,189
 - Rises sharply to 2012, 1,496
 - Rises gradually to 2013, 1,600
 - Falls sharply to 2014, 998
 - Falls gradually to 2015, 903
 - Falls gradually to 2016, 814
 - Rises sharply to 2017, 1,115
 - Falls sharply to 2018, 872
 - Falls sharply to 2019, 465

Investigative journalists research and report about fraud, corruption, public hazards, and more. The graph shows the number of investigative articles published in the

Albuquerque Journal newspaper from 2010 to 2019. According to an analyst, although the number of investigative articles published in this newspaper has varied significantly over the period shown, the number overall has fallen since 2010.

Which choice most effectively uses data from the graph to justify the underlined claim?

- A. The newspaper published approximately 1,000 investigative articles in 2010 and approximately 500 in 2019.
- B. The smallest annual number of investigative articles published in the newspaper during the period shown is approximately 1,600 in 2013.
- C. The greatest annual number of investigative articles published in the newspaper during the period shown is approximately 1,000 in 2017.
- D. The newspaper published approximately 1,000 investigative articles in 2010 and approximately 1,600 in 2013.

Comfort Ratings and Temperature-Adjustment Preferences from One Survey

Participant	Comfort rating	Preferred temperature adjustment
20	-2negative 2	Cooler
1	1	Cooler
21	1	Cooler

Nan Gao and her team conducted multiple surveys to determine participants' levels of comfort in a room where the temperature was regulated by a commercial climate control system. Participants filled out surveys several times a day to indicate their level of comfort on a scale from -3negative 3 (very cold) to +3positive 3 (very hot), with 0 indicating neutral (neither warm nor cool), and to indicate how they would prefer the temperature to be adjusted. The table shows three participants' responses in one of the surveys. According to the table, all three participants wanted the room to be cooler, _____blank

Which choice most effectively uses data from the table to complete the statement?

- A. and they each reported the same level of comfort.
- B. even though each participant's ratings varied throughout the day.
- C. but participant 20 reported feeling significantly colder than the other two participants did.
- D. but participant 1 reported feeling warmer than the other two participants did.

Depths at Which Four Deep-Sea Fish Species Live

Species	Depth below the ocean surface
Footballfish	200–1,000 meters
Southern stoplight loosejaw	500–2,000 meters
Black seadevil	250–2,000 meters
Bollons' rattail	300–800 meters

Some oceanic fish species live very deep underwater. Researchers collected data about the depths at which various species live.

Based on the information in the table, at what depth does the southern stoplight loosejaw live?

- A. More than 2,000 meters below the surface
- B. 150 to 400 meters below the surface
- C. 500 to 2,000 meters below the surface
- D. 250 to 500 meters below the surface

Estimated Impacts of Invasive Predator Species on Threatened Birds, Mammals, and Reptiles

Invasive predator species	Estimated impact on threatened birds	Estimated impact on threatened mammals	Estimated impact on threatened reptiles
Mongoose	0.66	0.75	0.44
Dogs	0.67	0.57	0.44
Cats	0.74	0.61	0.52

How harmful are invasive predators to threatened species? To find out, researchers assigned values from 0.25 to 1.00 to various harms caused by invasive predators. The researchers then used these values to estimate the extent to which several invasive predator species have harmfully impacted threatened birds, mammals, and reptiles. In the table, higher numbers represent greater impacts. The estimated impact of invasive mongooses on threatened mammals is _____blank

Which choice most effectively uses data from the table to complete the example?

- A. 0.35.
- B. 0.61.
- C. 0.75.
- D. 0.52.